
NEXGENBIOLAB

BIOINFORMATICS PROJECTS

AI-Based PROTEIN Structure Evaluation

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TASK 1 | AI-Based Structure Evaluation (AlphaFold3)

Objective: To demonstrate the ability to use AlphaFold3 to predict a protein structure and evaluate it against the experimental structure.

► Task Description

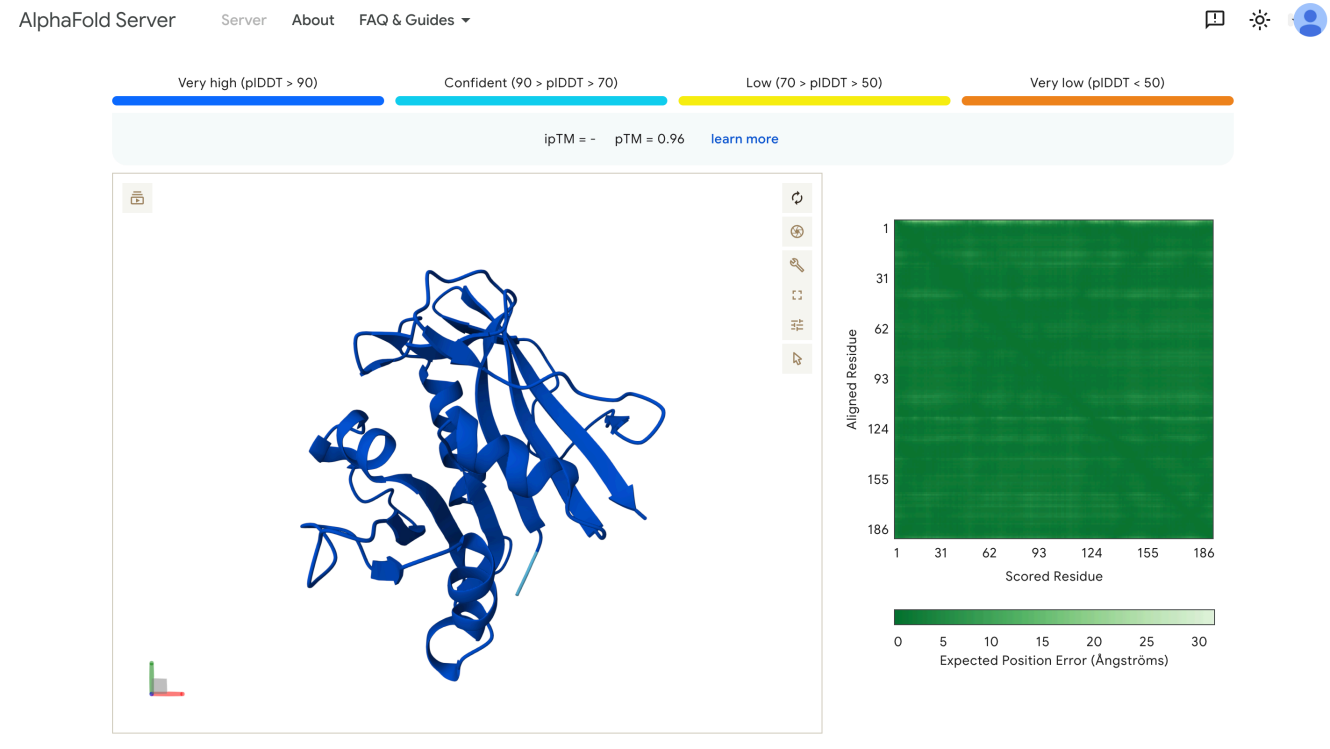
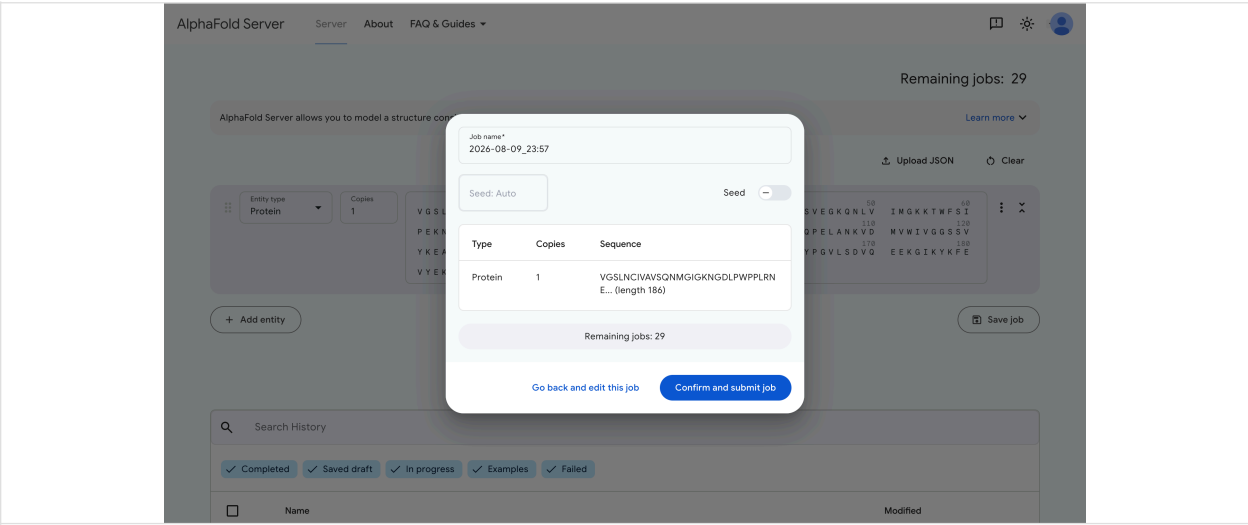
- 1 Go to the Protein Data Bank.
- 2 On the entry page, locate the FASTA sequence file. Download it (File → FASTA Sequence).
- 3 Open the FASTA file in a text editor and copy the sequence.
- 4 Go to AlphaFold3 (<https://alphafoldserver.com>) and create a new prediction job.
- 5 Paste the FASTA sequence into AlphaFold3 and submit the prediction.
- 6 Once complete, download the predicted structure.
- 7 Open both the predicted structure and your cleaned in PyMOL.
- 8 Use PyMOL's alignment function to align the predicted structure onto the experimental structure: type 'align predicted, experimental' in the PyMOL command line.
- 9 Note the RMSD value reported by PyMOL after alignment.

►FASTA Sequence

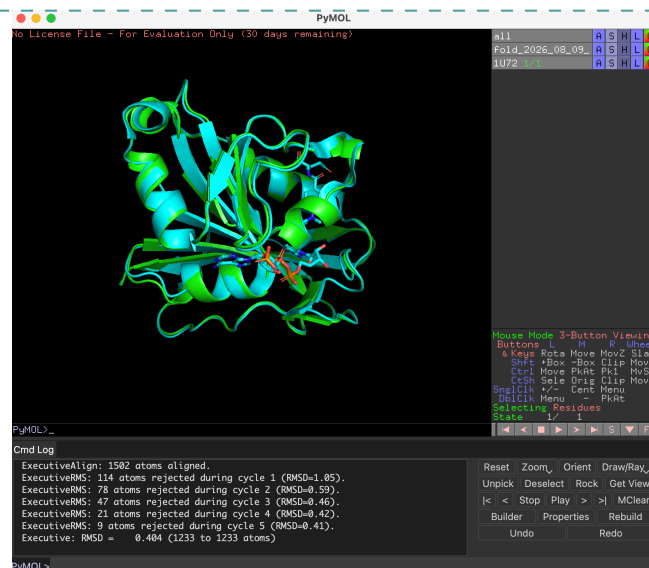
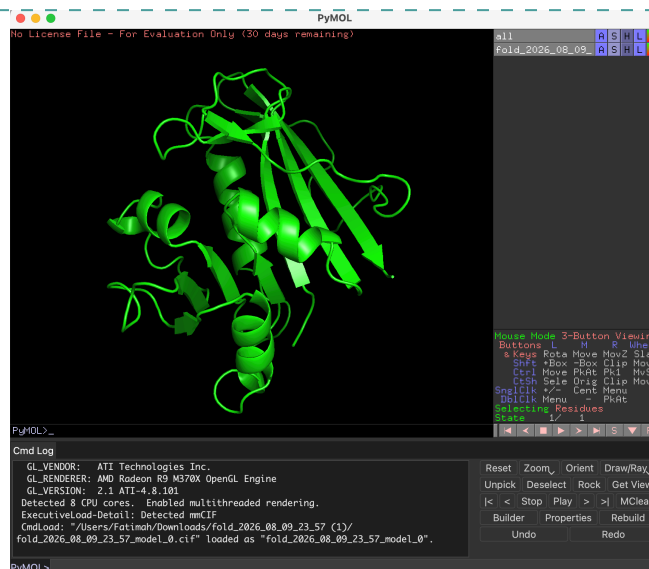
Paste the FASTA sequence in the box below.

```
>1U72_A | Paste sequence below:  
>1U72_1|Chain A|Dihydrofolate reductase|Homo sapiens (9606)  
VGS LNCIVAVSQNMGIGKNGDLPWPPLRNEFRYFQRM TTTSSVEGKQNLVIMGKKTWFSIPEKNRPLKGRINLVLSRELK  
EPPQGAHFLSRSLDDALKLTEQPELANKVDMVWIVGGSSVYKEAMNHPGHLKLFVTRIMQDFESDTFFPEIDLEKYKLLP  
EYPGVLSDVQEEKGIKYKFEVYEKND
```

► AlphaFold3 Website Screenshot



PyMOL Visualizations



► Alignment Quality Summary

Metric	Value	Interpretation
RMSD (Å)	0.404	$< 2 \text{ Å} = \text{excellent}$; $2\text{--}4 \text{ Å} = \text{good}$; $> 4 \text{ Å} = \text{poor}$
Residues Aligned	186 residues	Higher = better coverage
Overall Quality	Excellent agreement between the predicted and experimental structures, with a very low RMSD of 0.404 Å.	Excellent / Good / Acceptable / Poor

► **Written Justification**

How well does the AlphaFold3 predicted structure match the experimental structure

The AlphaFold3 predicted structure matches the experimental structure very well, with a low RMSD of 0.404 Å across the aligned region. This indicates a high degree of structural similarity between the predicted and experimentally determined structures.

